Beyond What Was Observed

Underlying Mechanisms Through Computational Materials Modeling

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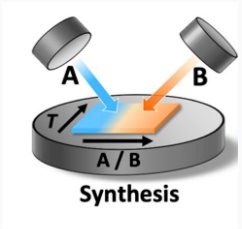
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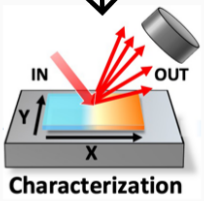


What Experiments Tell Us – and What They Do Not

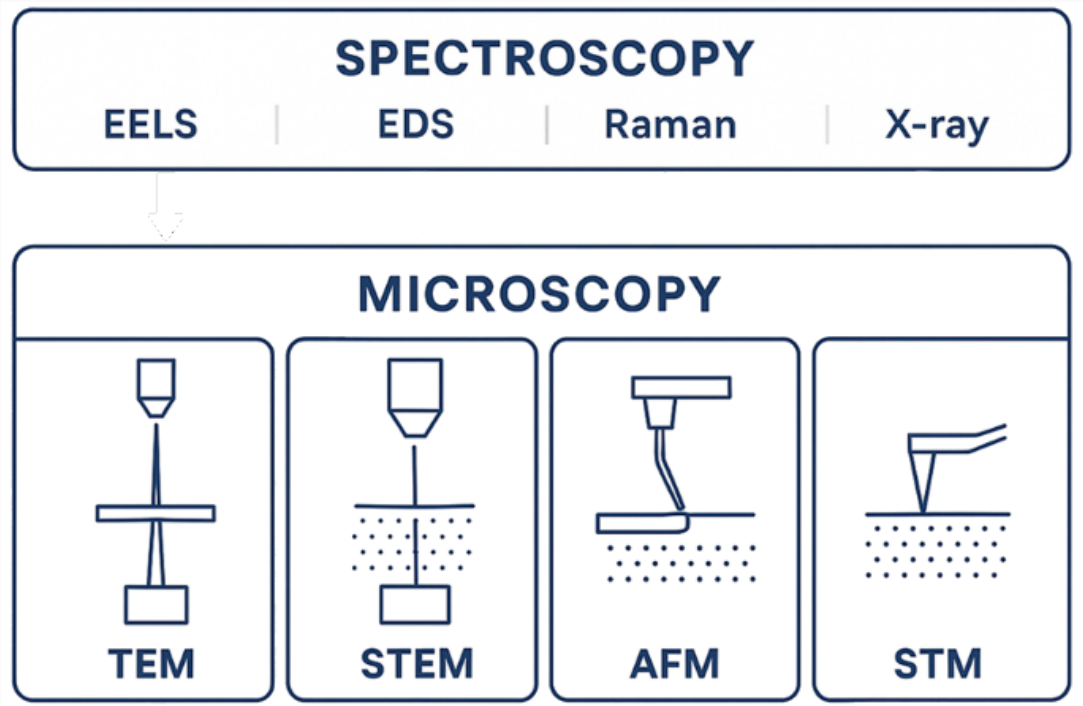
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Diffraction, microscopy and spectroscopy reveal structure and response, but the mechanism behind them is not always directly accessible.







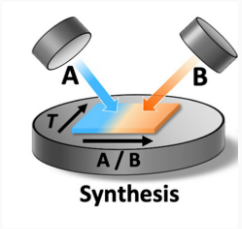


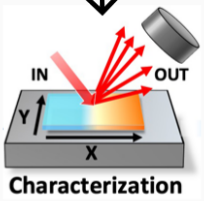
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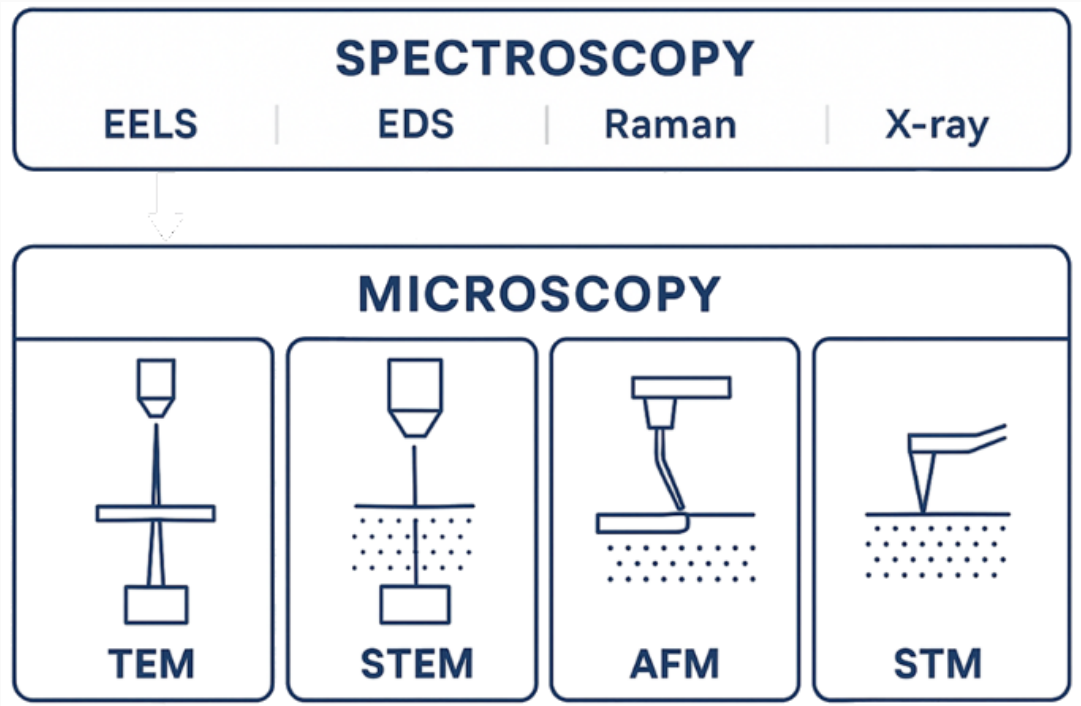
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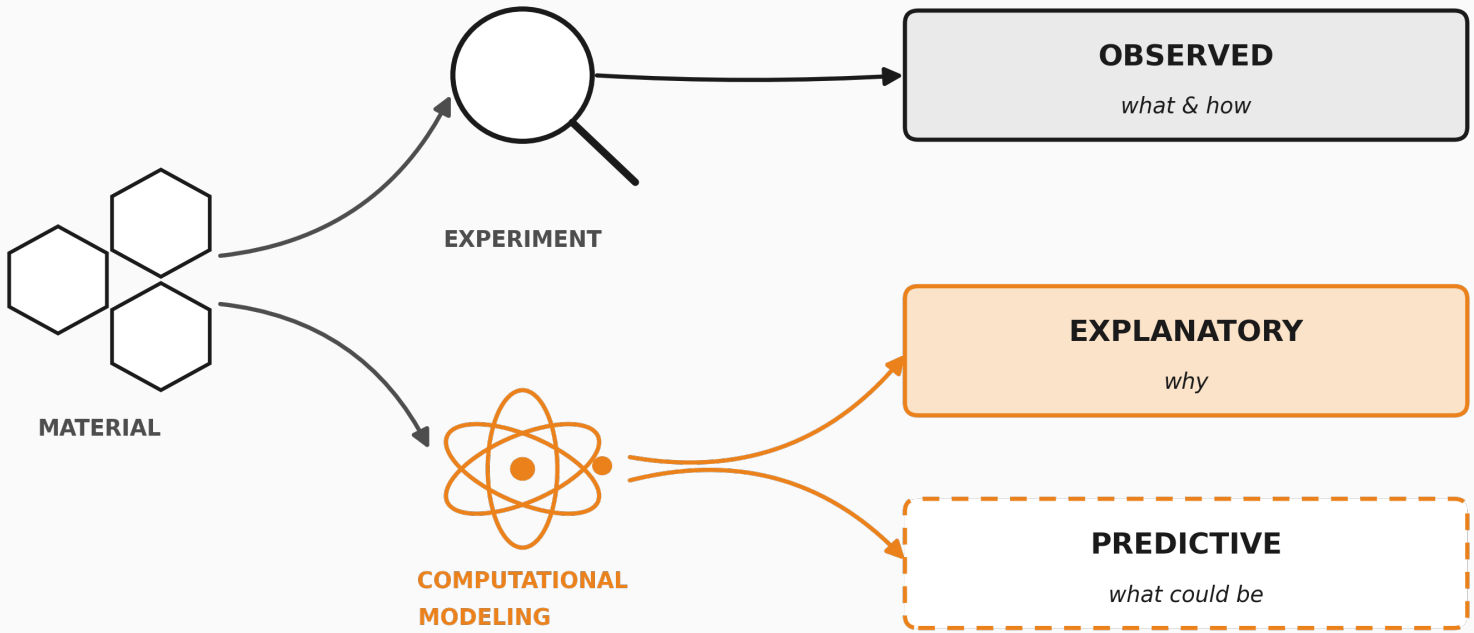
The gap

Between **what is observed** and **what governs the observation**, there is often room for a complementary approach.

Computational Modeling as a Second Lens

Access to quantities that are not directly measurable

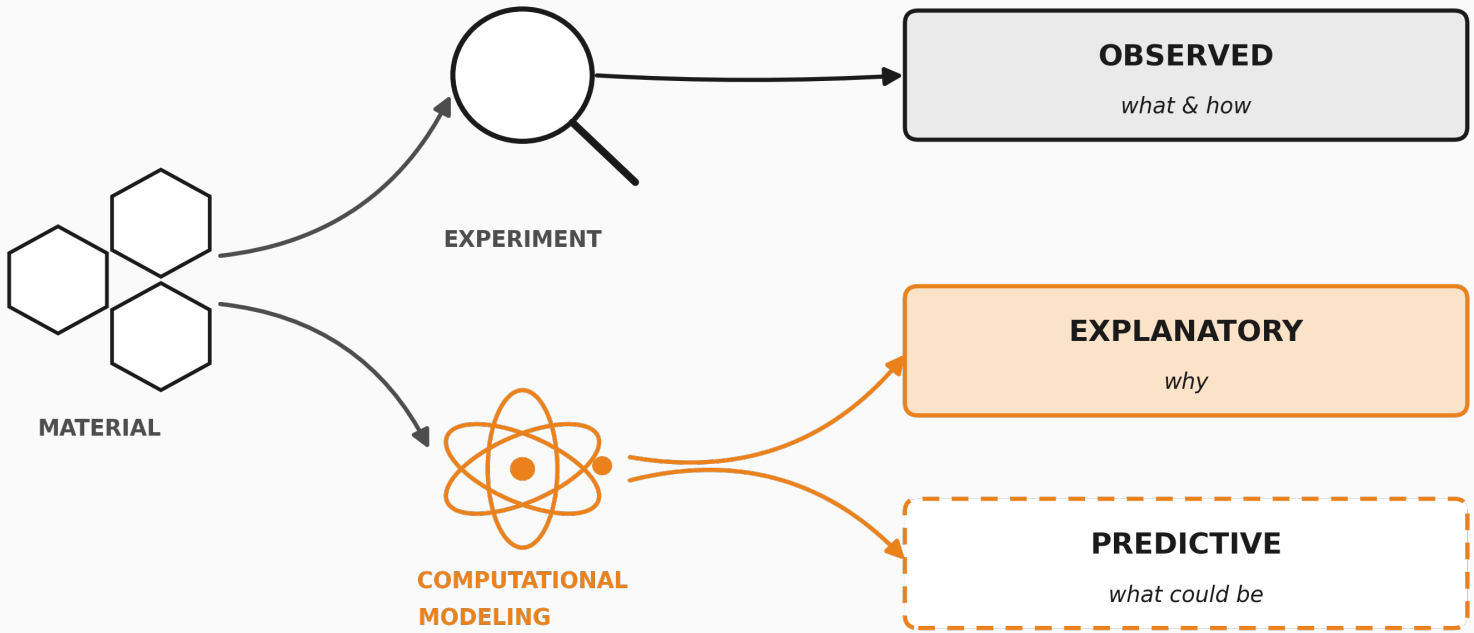
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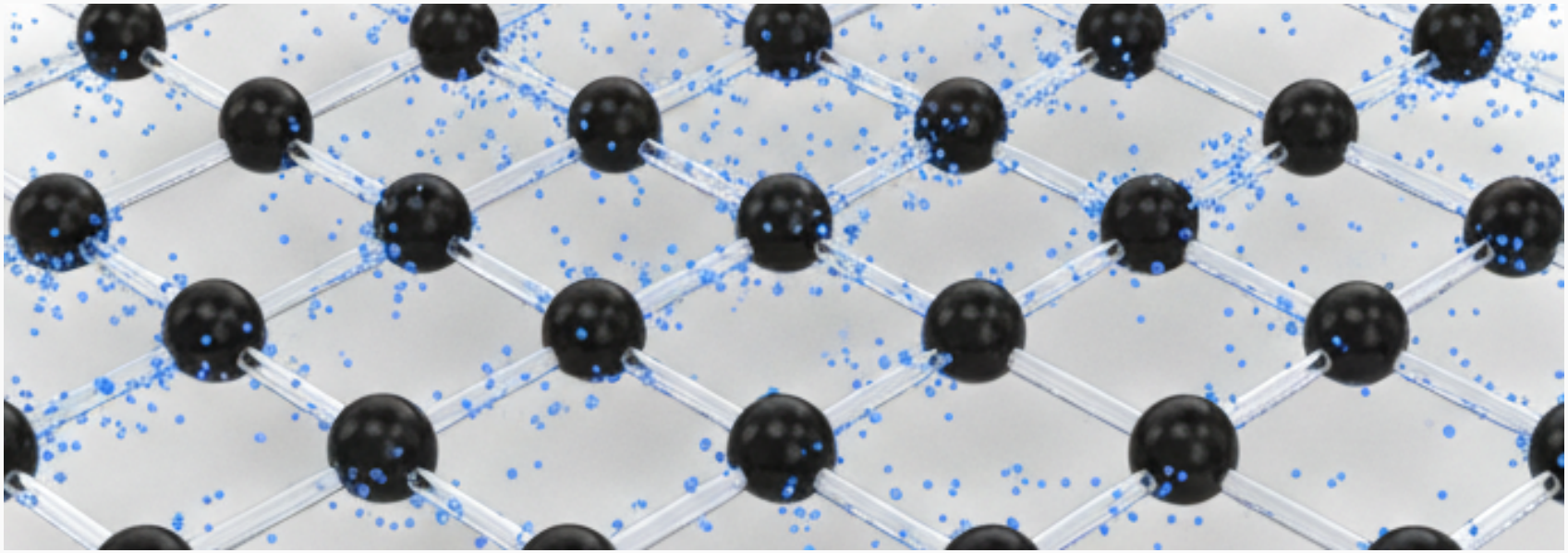


The question

What can we learn about a material beyond what has already been observed?

Methodology

The Many-Body Problem and the Hohenberg-Kohn Theorems



First Hohenberg-Kohn Theorem

The ground-state density $\rho_0(\mathbf{r})$ uniquely determines the external potential, and therefore the full many-body wave function:

$$|\psi_0\rangle = |\psi[\rho_0]\rangle$$

$3N$ variables are replaced by a function of 3 variables.

Second Hohenberg-Kohn Theorem

The energy functional is variational: for any trial density $\rho' \neq \rho_0$,

$$E[\rho_0] < E[\rho'] \quad \Longleftrightarrow \quad E_0 = \min E[\rho]$$

This gives us a practical route to the ground state.

Kohn-Sham mapping

Replace the interacting system by an **auxiliary non-interacting system** with the *same* ground-state density:

$$E[\rho] = T_s[\rho] + E_H[\rho] + E_{\text{xc}}[\rho] \qquad \{T_s + v_H[\rho] + v_{\text{xc}}[\rho]\}\,\phi_i^{\text{KS}} = \epsilon_i \phi_i^{\text{KS}}$$

All the difficulty is now contained in a single unknown term, $E_{\text{xc}}[\rho]$.



Electronic Density
- Interacting System
- Real System



Electronic Density
- Non-Interacting System
- Ficticius system



Electronic Density
- Non-Interacting System
- Ficticius system with the electronic density is the same of Real system

The Kohn-Sham Scheme and the Exchange-Correlation Problem

From jellium to real materials

- **LDA:** built from the homogeneous electron gas (jellium).
- **GGA (PBE):** adds the density gradient.
- **Hybrids (HSE06):** mix exact exchange, correcting the self-interaction error and improving band gaps.

Long-range dispersion

Semilocal functionals do not describe van der Waals interactions. **Grimme-type corrections (D3)** and non-local vdW functionals are essential for layered and porous systems – as we will see in the very first case study.

Implementation: Codes, Basis Sets and Tools

Plane waves – VASP

- Systematically convergent basis (cutoff energy), PAW potentials.
- Used for relaxations, band structures (PBE/HSE06) and phonons.

Localized orbitals – SIESTA

- Pseudo-atomic orbital basis (SZ, DZP): far cheaper for large supercells.
- Used for *ab initio* molecular dynamics, elastic constants and stress-strain curves.

Post-processing and many-body effects

- **Phonopy:** dynamical matrix and phonon dispersion.
- **Bader analysis:** quantifying charge transfer between atoms.
- **WanTiBEXOS:** Wannier tight-binding, Bethe-Salpeter equation, excitonic and optoelectronic properties. *(I am part of the*

The choice is part of the physics

Different questions require different tools: no single code answers structure, dynamics and optics at the same cost.

Case I: Layered materials are held together by van der Waals forces. Are they?

Transition Metal Dichalcogenides: From Bulk to 2D

A classical family of layered materials

TMDs (MX_2, with M a transition metal and X = S, Se, Te) are built from X–M–X trilayers stacked along the out-of-plane direction, each layer covalently bonded and weakly coupled to its neighbors.

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Why the interlayer coupling matters

- Exfoliation into monolayers is the standard route to 2D TMDs.
- The **exfoliation energy** sets how feasible this route is for a given compound.
- The textbook picture attributes the interlayer binding entirely to **van der Waals** forces.

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The question

Is the weak interlayer binding of TMDs really **only** a dispersion effect?

Bastos, C. M. O., Besse, R., Da Silva, J. L. F., Sipahi, G. M. (2019). *Phys. Rev. Materials* **3**, 044002.

A Systematic Survey of 27 Compounds

Chemical space explored

- **9 transition metals:** Ti, Zr, Hf (group IV); V, Nb, Ta (group V); Cr, Mo, W (group VI).
- **3 chalcogens:** S, Se, Te.
- **3 polymorphic phases:** 1T, 2H and 1T′ (distorted).

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Protocol

- Full geometry optimization including van der Waals corrections; lowest-energy phase identified for each composition.
- Magnetic ordering tested (ferro- and antiferromagnetic solutions found for V-group and Cr compounds).
- Elastic constants and **Born stability criteria** evaluated for all systems.
- Good agreement with experimental data for the already synthesized compositions – including compounds **not yet synthesized**.

Bastos, C. M. O., Besse, R., Da Silva, J. L. F., Sipahi, G. M. (2019). *Phys. Rev. Materials* **3**, 044002.

Structural and Electronic Trends

The *d*-band tells the story

The **occupation of the metal *d* band** determines whether the compound behaves as a metal, a semimetal or a semiconductor.

Band gap trends

- Heavier chalcogen $\rightarrow$ **narrower** gap (the X-*p* states forming the VBM are pushed up in energy).
- Heavier transition metal $\rightarrow$ **wider** gap (e.g. $TiS_2 < ZrS_2 < HfS_2$ in the 1T phase), as the *d* orbitals become more localized and move away from E_F.

Functional matters

Semilocal PBE severely underestimates (or closes) the gaps; HSE06 restores agreement with the available experimental values.

Bastos, C. M. O., Besse, R., Da Silva, J. L. F., Sipahi, G. M. (2019). *Phys. Rev. Materials* **3**, 044002.

Beyond the van der Waals Picture

What we found

The exfoliation energy does **not** follow a simple dispersion trend. It correlates with the **charge transfer within each layer**, quantified by Bader analysis.

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Why this is beyond what was observed

Exfoliation energies can be inferred experimentally, but the electrostatic origin cannot: the measurement returns a number, while the calculation returns the **reason** behind it – and a chemical handle (charge transfer) to tune it.

Bastos, C. M. O., Besse, R., Da Silva, J. L. F., Sipahi, G. M. (2019). *Phys. Rev. Materials* **3**, 044002.

Take-Home Message – and the Next Question

What we learned

- A property universally attributed to weak dispersion forces has a clear **electrostatic contribution**.
- Most of the 27 TMDs are predicted to be easy to exfoliate, including compositions not yet synthesized.
- Modeling explained the **mechanism behind** a well-known experimental observation.

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Bridge

Here we explained a **known** structure.
But what if the lattice itself is the exotic ingredient?

Case II: A single-element kagome lattice. Stable or not?

Kagome Lattices in Noble-Metal Monolayers

Why kagome

The corner-sharing triangular network hosts **flat bands**, **Dirac cones** and geometric frustration – key ingredients for topological and strongly correlated phenomena.

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Why noble metals

- Kagome physics is usually realized in complex multi-element compounds.
- The recent synthesis of **goldene** – a true single-atom-thick gold layer – shows that elemental 2D metals are now experimentally accessible.
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This study

Free-standing **Cu, Ag and Au** kagome monolayers: are they stable?

Bastos, C. M. O., dos Santos, E. J. A., Laranjeira, J. A. dos S., Lima, K. A. L., Dias, A. C., Galvão, D. S., Ribeiro Jr., L. A. (2026). *Chem. Phys. Lett.* **887**, 142676.

Mechanically Stable – and Unusually Soft

Born criteria for 2D lattices

$$C_{11} > 0 \qquad \frac{C_{11} - C_{12}}{2} > 0 \qquad C_{11} > |C_{12}|$$

All three kagome monolayers (Cu, Ag, Au) satisfy the criteria: they are mechanically stable.

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Reduced coordination, reduced stiffness

The open kagome topology gives a much lower in-plane stiffness than dense 2D metals or graphene – the lattice pays a mechanical price for its geometry.

Gold stands out

Au shows the **largest Young's modulus** of the three, reflecting stronger interatomic bonding driven by **relativistic effects** – the same effects behind the color and chemistry of gold.

Bastos, C. M. O. *et al.* (2026). *Chem. Phys. Lett.* **887**, 142676.

Mechanical Stability Is Not Enough

Phonon dispersion of the unstrained lattices

All three systems display **soft (imaginary) modes** in the free-standing, unstrained configuration: mechanically stable, yet dynamically unstable.

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Strain as a stabilizing knob

- A moderate **biaxial tensile strain** fully suppresses the soft modes in **Ag and Au**.
- In **Cu**, residual instabilities persist even under strain.
- This points directly to **substrate-assisted growth** as a synthetic route.

Bastos, C. M. O. *et al.* (2026). *Chem. Phys. Lett.* **887**, 142676.

Beyond the Static Picture: Gold at Room Temperature

AIMD at 50 K and 300 K

- **Cu:** kagome phase collapses even at low temperature.
- **Ag:** metastable at 50 K; at 300 K the energy drops by $\sim$32 meV/atom as the system converts to a trigonal phase.
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The underlying mechanism: phase coexistence

Gold does not simply transform. Kagome-like and trigonal-like motifs **coexist** within the same lattice at 300 K, with average energies differing by only ∼26 meV/atom. The two configurations sit on a **shallow energy landscape**, and both can be locally stabilized.

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Why this is beyond what was observed

A static structural characterization would return a single averaged structure. The competition between two nearly degenerate motifs is a **dynamical, local** effect – visible only when we let the system evolve in time.

Take-Home Message – and the Next Question

What we learned

- Three levels of stability – mechanical, dynamical and thermal – give **three different answers** for the same lattice.
- Lattice topology, atomic size and relativistic bonding jointly decide the outcome.
- For gold, the model reveals a regime of **structural competition** inaccessible to static probes.
- The result provides a framework for **strain- and substrate-assisted** realization.

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Bridge

So far, two-dimensional lattices.
What happens when we move into a fully three-dimensional network?

Case III: A 3D carbon allotrope was synthesized. Is that the whole story?

The Quest for New 3D Carbon Allotropes

A crowded landscape in low dimensions

Fullerenes (0D), nanotubes (1D), graphene, graphyne, biphenylene and monolayer fullerene networks (2D) have all been synthesized in the last decades.

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The opportunity

The structure has been made. Its **internal organization**, mechanical response and electronic character remained largely unexplored.

Bastos, C. M. de O., dos Santos, E. J. A., Alves, R. A. F., Cavalheiro Dias, A., Ribeiro Junior, L. A., Galvão, D. S. (2025). *ACS Omega* **10**(39), 46065–46070.

Diamondiynes: Entangled and Interlocked Lattices

Mixed hybridization

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The expanded lattice can accommodate **two or more independent sublattices**, threaded through one another without any chemical bond between them – **entangled and interlocked** networks (NI, 2F, 2F-USY, 3F, 4F).

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The question this raises

If the sublattices are not bonded to each other, **what holds them in place?**

Bastos, C. M. de O. *et al.* (2025). *ACS Omega* **10**(39), 46065–46070.

Stable, Semiconducting and Tough

Thermal stability

AIMD confirms that **all structures remain stable at 300 K and 1000 K**, with no bond breaking or lattice collapse.

Electronic properties

All diamondiyne structures are **wide-gap semiconductors**, attractive for electronic and optoelectronic applications.

Mechanical properties

The combination of **flexibility and high strength** makes them promising as impact-absorbing materials.

Bastos, C. M. de O. *et al.* (2025). *ACS Omega* **10**(39), 46065–46070.

Beyond the Static Structure: The Lattice Moves

What the dynamics reveal

- C–C bonds **within** each sublattice (single and triple) stay stable even at 1000 K: the chemistry is intact.
- Distances between atoms belonging to **different** sublattices fluctuate widely and randomly.

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Why this is beyond what was observed

Diffraction resolves the **average** positions of the atoms. The relative mobility of the sublattices is invisible in a static structure – it only emerges once the lattice is allowed to evolve in time.

Bastos, C. M. de O. *et al.* (2025). *ACS Omega* **10**(39), 46065–46070.

Take-Home Message – and the Next Question

What we learned

- A synthesized material carried a property that the synthesis itself did not reveal.
- Mechanical interlocking at the **atomic scale** – a molecular-machine motif inside a bulk carbon solid.
- Modeling did not confirm the experiment: it **extended** it.

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Bridge

Until now, we looked at materials that **already exist**.
Can we say something useful about one that **does not exist yet**?

Case IV: Only one member of the family has been synthesized. Is it the best one?

Chalcogenophene-Modified Graphyne

Graphyne: a tunable carbon platform

γ-graphyne combines *sp* and sp^2 carbons, forming a porous π-conjugated semiconductor. Its properties can be tailored by **replacing the benzenoid rings** with heterocyclic units.

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A route chemists already know

Sonogashira cross-coupling of 2,3,4,5-tetrabromo-chalcogenophene precursors with calcium carbide as the acetylene source gives access to the whole family:

furan (fu-GY) · thiophene (thp-GY) · selenophene (sep-GY) · tellurophene (tep-GY)

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The starting point

Only **thp-GY** has been realized experimentally (reported as sulfur-doped graphyne). The other three members remain hypothetical.

Bastos, C. M. O., Oliveira, I. da S., Batista, A. L. de O., Silva, A. M. A., Villegas, C. E. P., Dias, A. C., Marinho Jr., E., Ribeiro Júnior, L. A. (2026). *J. Mater. Chem. A*, DOI: 10.1039/D6TA05506K.

All Four Members Are Stable

Structural

All monolayers relax to **strictly planar** lattices without symmetry constraints. The lattice parameter grows from 6.22 Å (fu-GY) to 6.34 Å (tep-GY), following the chalcogen atomic radius.

Energetic

Cohesive energies from -6.86 to -6.48 eV/atom, comparable to γ-graphyne (~ -7.21) and graphene (~ -7.92).

Mechanical

Born criteria satisfied; $C_{11} > 217$ N/m despite the porous architecture. Poisson's ratio (0.14–0.17) close to graphene.

Dynamical and thermal

No imaginary phonon modes; AIMD at 300 K shows no bond breaking or out-of-plane rippling over 5 ps.

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Consequence

Stability is not the discriminating factor. **The chalcogen is.**

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The Chalcogen as a Chemical Knob

Charge transfer changes sign along the group

Bader analysis: oxygen is a strong **acceptor** ($-1.51e$), while S, Se and Te act as progressively stronger **donors** ($+0.44e$, $+0.56e$, $+2.10e$).

Consequences for the band structure

- Indirect gaps between **0.93 and 1.31 eV** – ideal for visible and near-infrared harvesting.
- Work function tuned from 4.91 eV (fu-GY) to 4.76 eV (tep-GY).
- Electron affinity rising from 3.97 to 4.18 eV.
- Heavier chalcogens push the system towards **n-type** behavior.

The Chalcogen as a Chemical Knob

Charge transfer changes sign along the group

Bader analysis: oxygen is a strong **acceptor** ($-1.51e$), while S, Se and Te act as progressively stronger **donors** ($+0.44e$, $+0.56e$, $+2.10e$).

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The mechanism

A single atomic substitution rewires the electronic structure through **charge transfer**, not through geometry alone.

Excitonic and Optical Response

Beyond the independent-particle picture

Optical properties obtained by solving the **Bethe-Salpeter equation** on a Wannier tight-binding Hamiltonian extracted at the HSE06 level (WanTiBEXOS).

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Strongly bound excitons

Binding energies of **328–457 meV**, at the upper end of the range reported for 2D semiconductors – a direct consequence of reduced dielectric screening in the monolayer limit.

Absorption

Strong absorption across IR, visible and UV, with marked optical anisotropy. Going down the chalcogen group shifts the excitonic resonances, giving **optimal visible-range absorption for sep-GY**.

Bastos, C. M. O. *et al.* (2026). *J. Mater. Chem. A*, DOI: 10.1039/D6TA05506K.

Beyond What Was Synthesized

Simulated single-junction solar cell

Active layer between TiO_2 (ETL) and P3HT (HTL), with ITO and an Ag back reflector. Band alignment is favorable for charge extraction in all four monolayers.

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Efficiency follows **Se** > **O** > **S** > **Te**.

	PCE (radiative)	PCE (non-radiative)
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The twist

The only member that chemists have actually made is the **worst** candidate of the family. Here modeling stops explaining the past and starts **pointing at the next experiment** – together with a verification protocol: HRTEM/STM, XPS/Raman, ARPES/UPS and UV-Vis-NIR absorption.

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Conclusion

Four Materials, One Pattern

Transition metal dichalcogenides

Interlayer binding is not only van der Waals: **charge transfer drives Coulomb repulsion** between chalcogen planes, governing exfoliation.

Kagome noble-metal monolayers

Mechanical stability is not dynamical stability: gold displays a **coexistence of kagome and trigonal motifs** at room temperature.

Diamondiynes

A synthesized 3D carbon allotrope whose interlocked sublattices are held **topologically rather than chemically** – and are free to move.

Chalcogenophene-modified graphyne

The **unsynthesized** selenophene variant outperforms the synthesized thiophene one for photovoltaics, by a factor larger than two.

Beyond What Was Observed

Two directions, one tool

- **Looking back:** recovering the mechanism behind properties that have already been measured – electrostatics behind exfoliation, dynamics behind an averaged structure.
- **Looking forward:** evaluating materials that do not yet exist, and identifying which one is worth the synthetic effort.

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Not a confirmation step after the experiment, and not a replacement for it. The calculation and the bench answer **different questions** about the same material.

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Experiments show us what a material **is**.
Modeling tells us **why** – and what it **could be**.

Acknowledgments

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Thank You & Let's Connect!

We're always open to new collaborations...

...especially if they involve discussing physics over a good cup of coffee!



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Thank you!